

1. Evaluate $\int x \cos(4x) dx$

2. Evaluate $\int \sin^3 x \cos^6 x dx$

Formula Sheet

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1, & \sin^2 x &= 1 - \cos^2 x, & \cos^2 x &= 1 - \sin^2 x \\ \sec^2 x &= \tan^2 x + 1, & \tan^2 x &= \sec^2 x - 1\end{aligned}$$